

SUBSET SUBSEQUENCES

≠
STRING

QUESTIONS

Q11 str = b a c c a d

ans = b c c d

Removing a specific character

M-1 Pass ans string in argument

M-2 Create the ans variable in fun body.

M-1

Ans / String

t=0

" " / "b a c c a d"

↓ b == a X Add

" b " / " a c c a d"

↓ a == a ✓ skip

" b " / " c c a d"

↓ c == a X add

" bc " / " c a d"

↓ c == a X add

" bcc " / " a d"

↓ a == a ✓ skip

" bcc " / " d"

↓ d == a X add

" bcc d " / " "

skip
bccd

OBSERVATION

return

① Original string Empty → this found

② add & skip game

M-2

this variable in body of fun

(b a c c a d)

[ch = b +]

[ch = " " +]

(c c a d)

[ch = c +]

(c a d)

[ch = c +]

(a d)

Hey you are going

above please take me

d

ch = a +]

(d)

[ch = d]

p → Processed → ans

up → unprocessed → original

Static void skip (String p, String up)

if (up.isEmpty()) {

System.out.println(p);

return;

// base case

M-2

char ch = up.charAt(0);

if (ch == 'a') {

skip (p, up.substring(1);

}

else {

skip (p + ch, up.substring(1);

}

}

SUBSETS

P & C

subsets $\rightarrow$ Non-adjacent collection

$[3, 5, 9] \rightarrow [3], [3, 5], [3, 9],$
 $[3, 5, 9], [], \dots$

Rec & Memo

Q Create all the subset

Str = "abc"

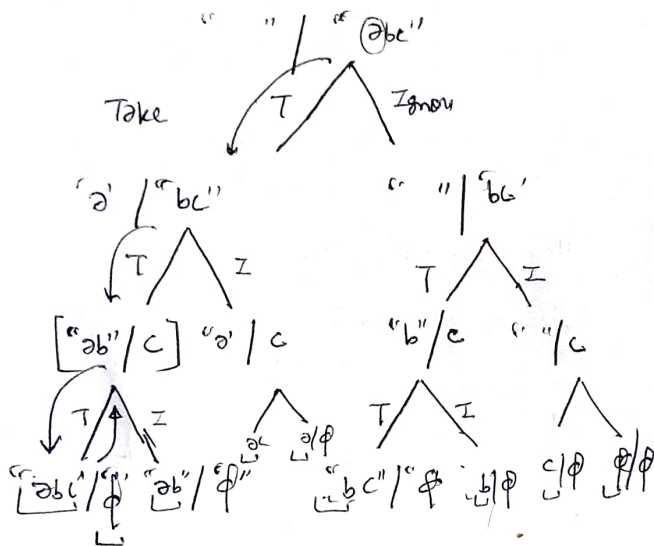
ans = ["a", "b", "c", "ab", "ac",

bc

ordering can't change ✓

VVI

SUBSET PATTERN



OBSERVATION

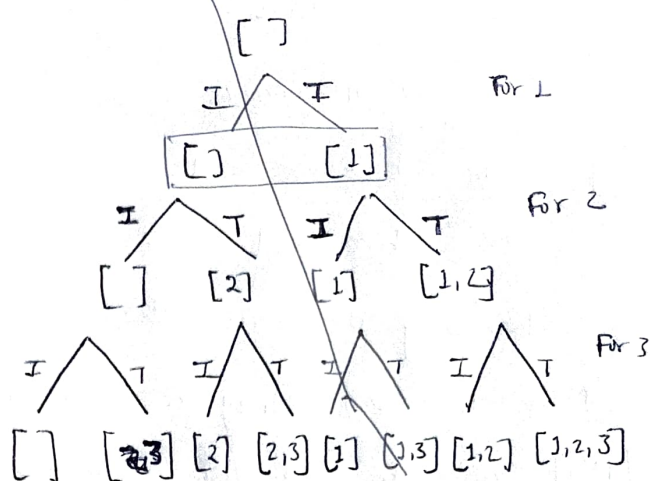
UP $\rightarrow$ empty // Base Case

Ascii

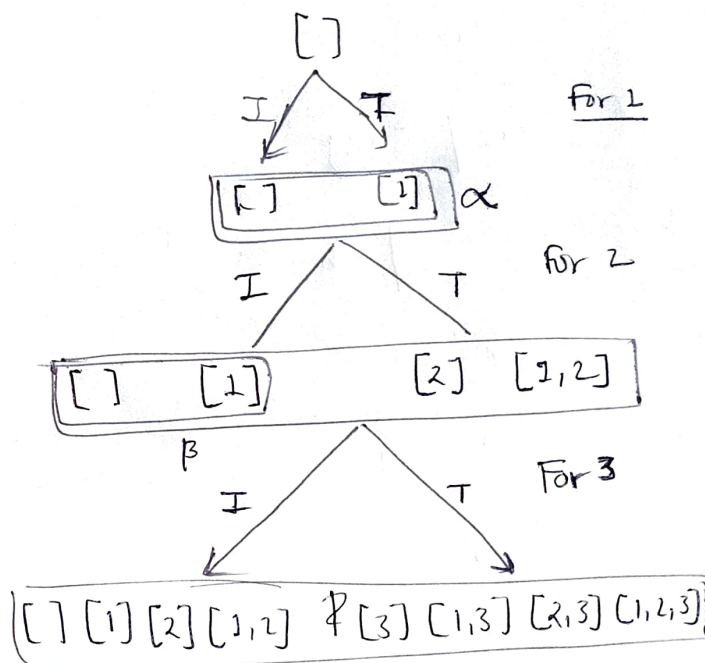
(k+1)

LOOPS

nums = [1, 2, 3]



LOOPS



$\alpha = \beta$

$[], [], [], [], [] \rightarrow$

for(i=0; i<=2; i++)

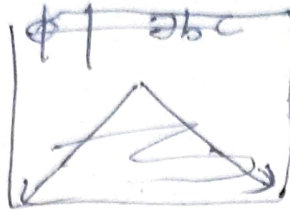
— b — a —
 ↙ ↘
 [0,0] [0,1] that
 +ch
 + (j, end)

PERMUTATIONS

Str = abc

permutations = [abc, bac, cab, bca, acb, ---]

Apply PFC



No of places in which we can add

$\phi = ""$

$\phi / a | bc$

— a —

$a | b | c$

R L

ab/c

ba/c

— a — b —

— b — a —

R M L

R M L

c ab / ϕ a cb / ϕ bac / ϕ c ab / ϕ acb / ϕ abc / ϕ

Observation

- At every recursive level there is a variable no of recursive call
- No of calls dependence on size of Process

Size of Process Calls

0

1

1

2

2

3

No of Recursive calls = size of Process + 1